Introduction to Economics

Class 8 — Surplus, efficiency, taxes and subsidies

Pavel Kocourek

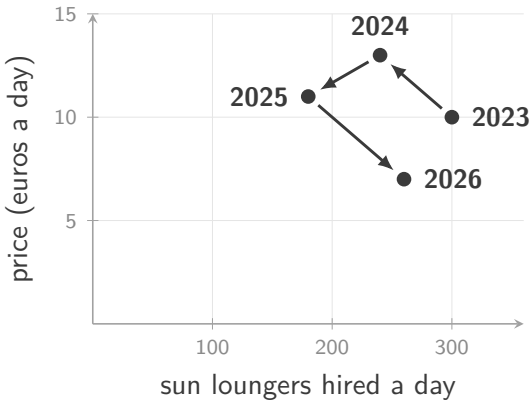
Universitat de Barcelona · Group A6

19 October 2026

Plan for today

Warm-up: which curve moved?
Bread, milk, eggs: VAT down to zero
What buyers gain: consumer surplus
What sellers gain: producer surplus
In pairs: menús del día
Total surplus: efficient, but fair?
A tax: a wedge between two prices
Break
Who pays the tax office; where the money goes
In pairs: a tax on menús
Who bears a tax: elasticity decides; in pairs
A subsidy; VAT; the bread answered
In one line; before Thursday 22 October

Warm-up: two dots — which curve moved?



Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
2. 2024 → 2025
3. 2025 → 2026

In pairs, two minutes; then hands per arrow.

1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %



bread, milk, cheese, eggs, fruit, vegetables

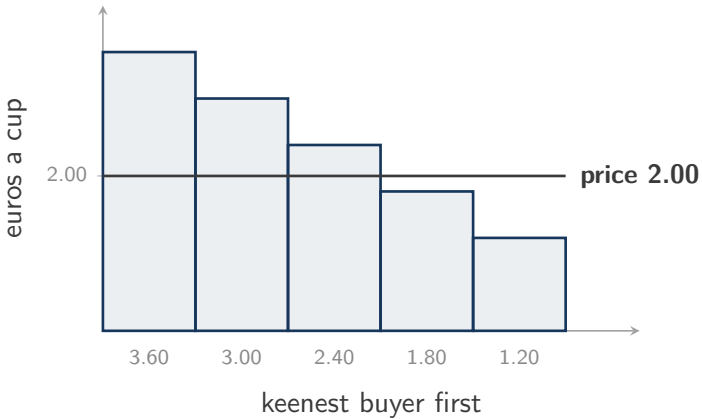


olive and seed oils, pasta

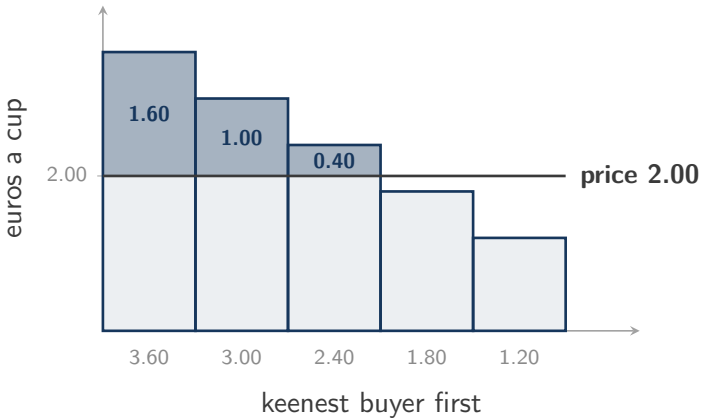
A 1.04 € loaf: 4 cents of VAT.
How many reached the shopper?

Real Decreto-ley 20/2022 (BOE, 28 December 2022), in force 1 January 2023.

Willingness to pay: each step of demand is one buyer

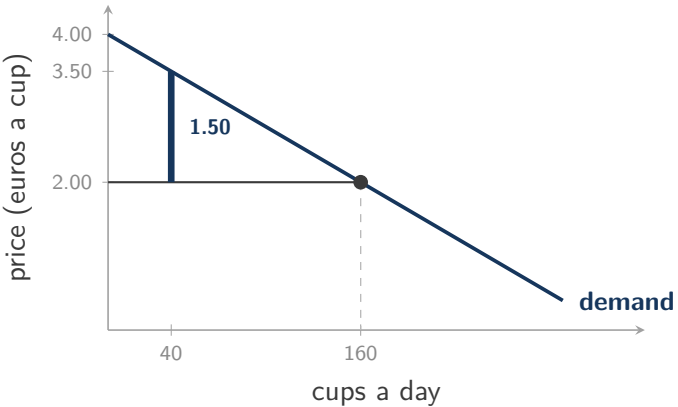


Willingness to pay: each step of demand is one buyer

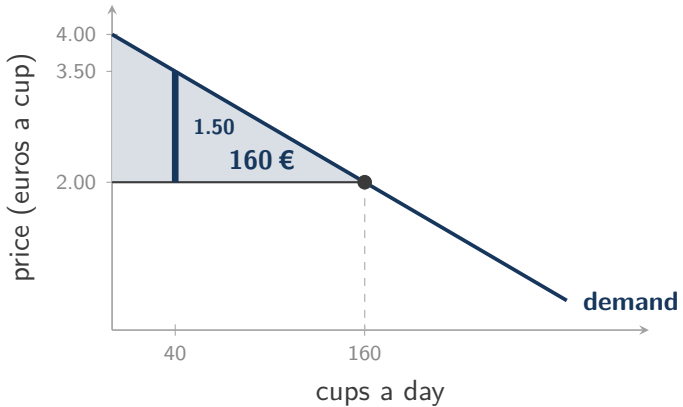


Consumer surplus = willingness to pay − price: $1.60 + 1.00 + 0.40 = 3.00\,€$.

Consumer surplus: under demand, above the price — 160 € a day

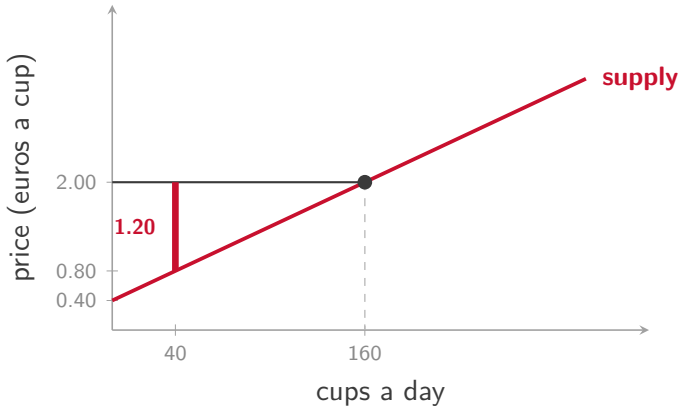


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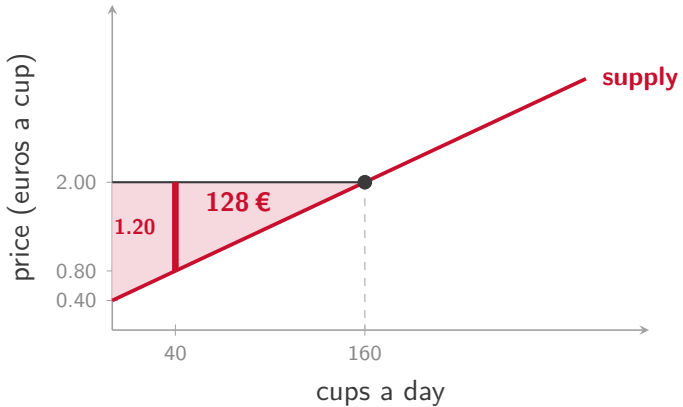


½ × 160 cups × 2.00 € = 160 euros a day: every buyer's bargain, added up.

Producer surplus: above supply, below the price — 128 € a day

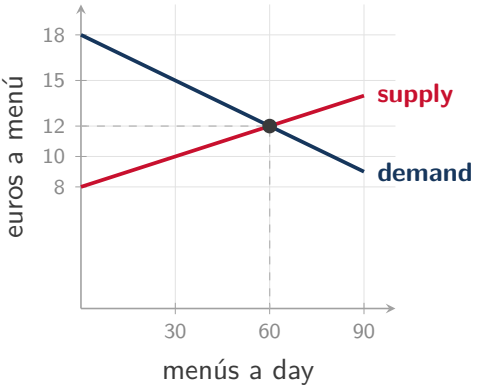


Producer surplus: above supply, below the price — 128 € a day



½ × 160 cups × 1.60 € = 128 euros a day: what the bars get above their costs.

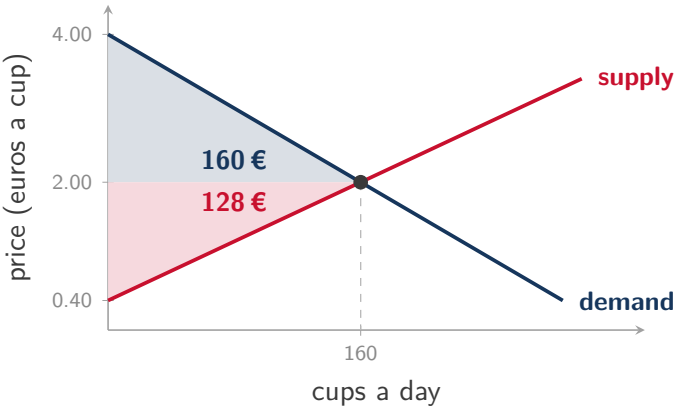
In pairs, four minutes: read the surplus off the graph



Menús del día on one street

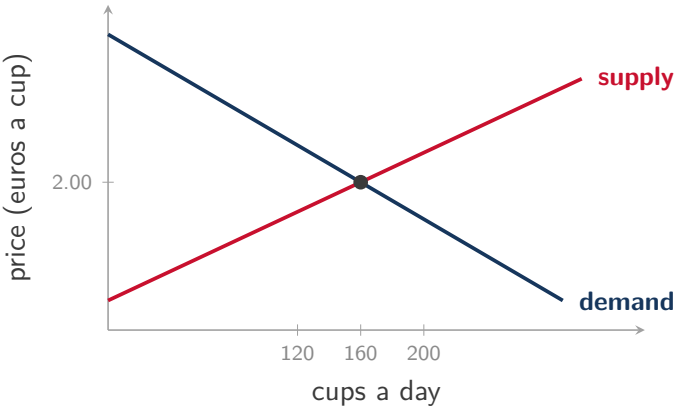
1. Consumer surplus at 12 €?
2. Producer surplus at 12 €?
3. The 30th menú: worth? costs?

Total surplus: buyers' 160 plus the bars' 128 — 288 euros a day

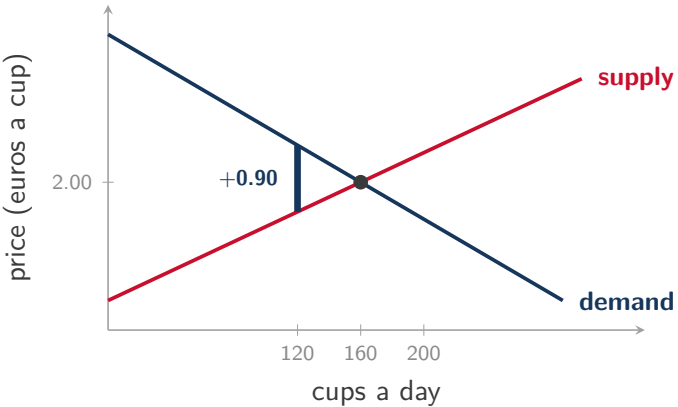


Total surplus = value to buyers − cost to sellers = 480 − 192 = 288 euros a day.

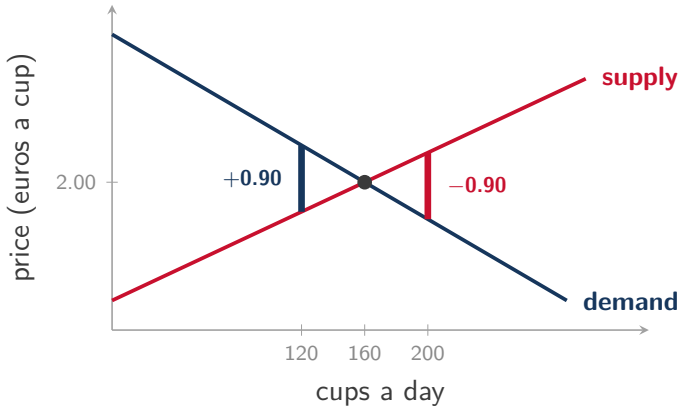
Stop at 160: before it, cups worth more than they cost; after it, less



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Stop at 160: before it, cups worth more than they cost; after it, less



Below 160: surplus left on the table. Above: cups that cost more than they are worth.

Just right: the market's 160 cups give the largest total surplus

- The cups go to those who value them most
- They are made by the bars that make them cheapest
- Every cup worth more than its cost, and no other

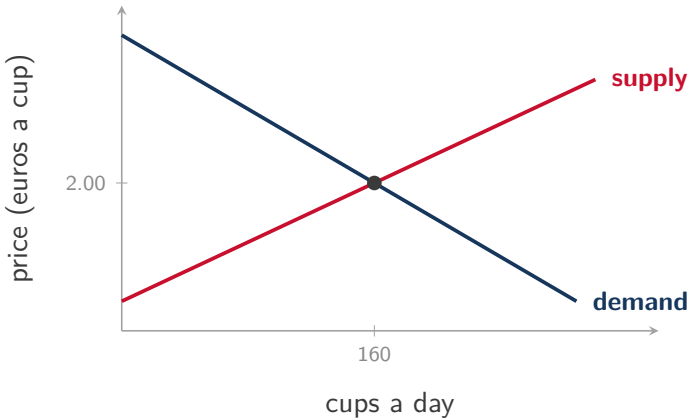
That is what economists call **efficient**, as long as no trade affects others (Class 10).

Efficient is not the same as fair

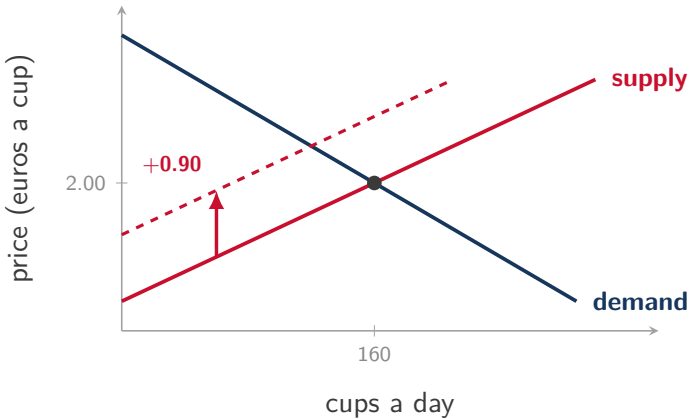
- Efficiency: is the cake as big as it can be?
- Equity: is it shared fairly?
- Willingness to pay depends on income

Efficiency is a positive question; fairness a normative one (Class 4).

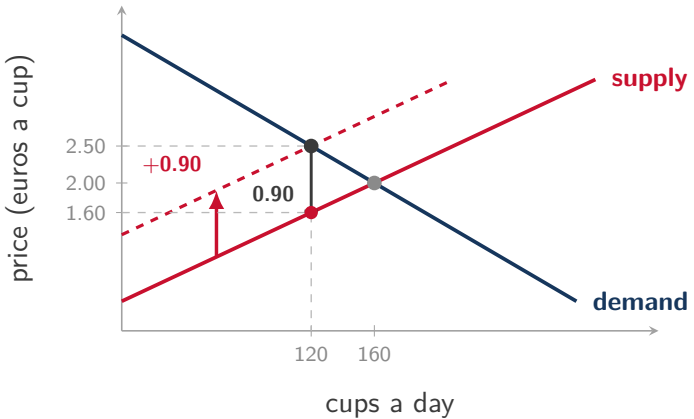
A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



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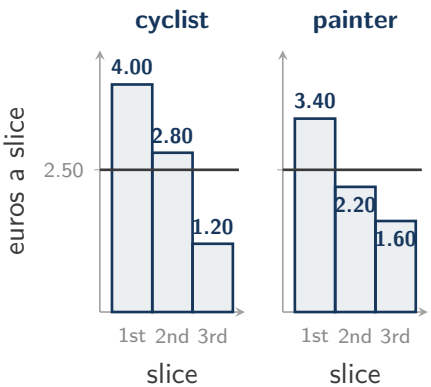


A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



Buyers pay 0.50 more, the bars keep 0.40 less: a tax is a wedge between two prices.

Market demand from two staircases (optional)



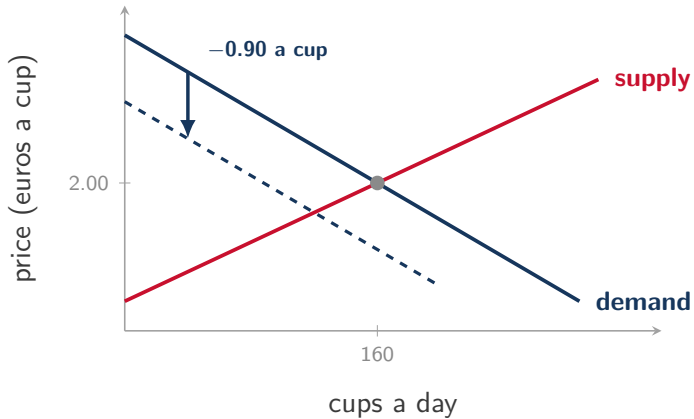
Pizza slices: two buyers

1. At 2.50: slices bought in all?
2. Their consumer surplus at 2.50?
3. Slices bought in all at 2.00?

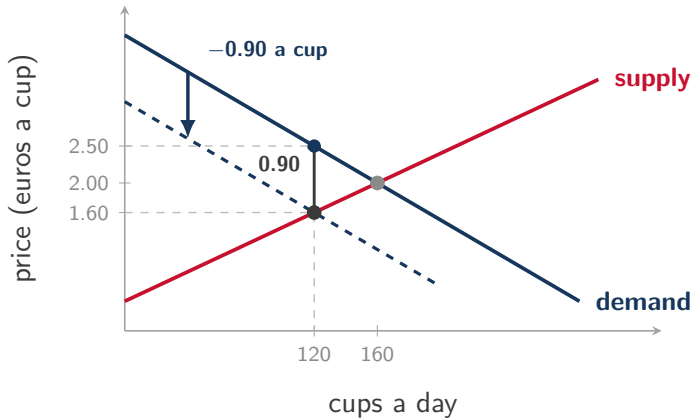
Break

Ten minutes

Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

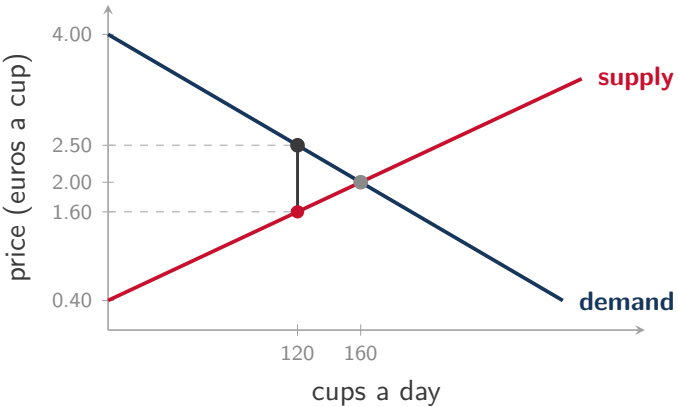


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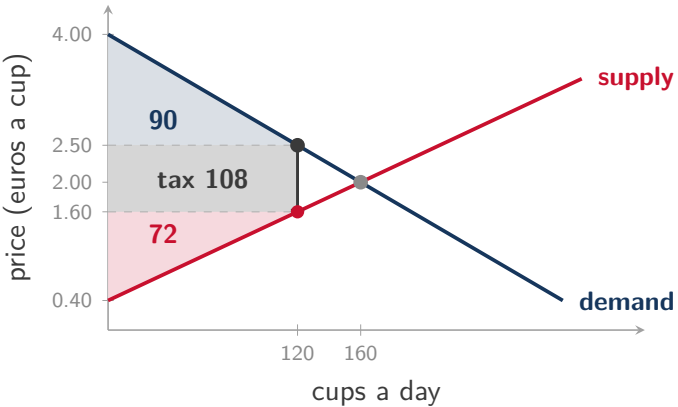


Who hands the 0.90 to the tax office does not matter: the wedge does.

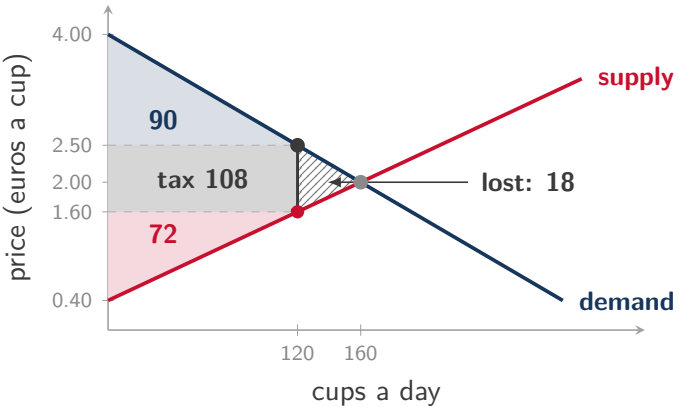
With the tax: buyers 90, bars 72, the state 108 — and 18 lost



With the tax: buyers 90, bars 72, the state 108 — and 18 lost



With the tax: buyers 90, bars 72, the state 108 — and 18 lost



Buyers lose 70, the bars 56; the state gets 108. The lost 18: the **deadweight loss**, Thursday.

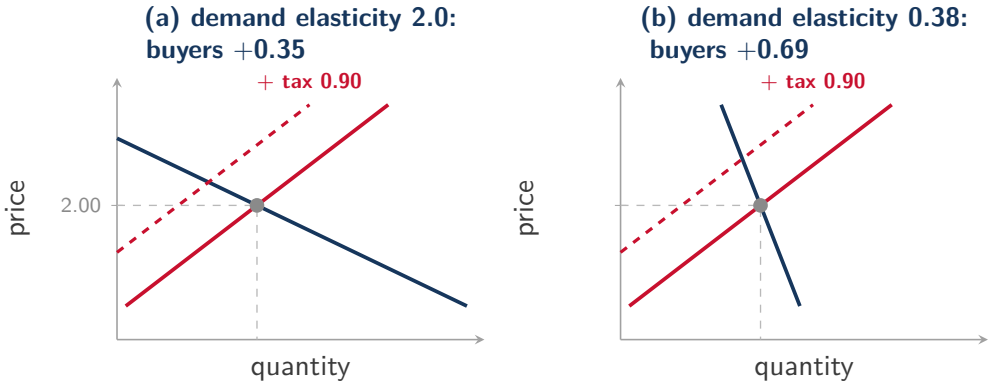
In pairs, four minutes: a tax of 2.50 on every menú

Same street, same menús

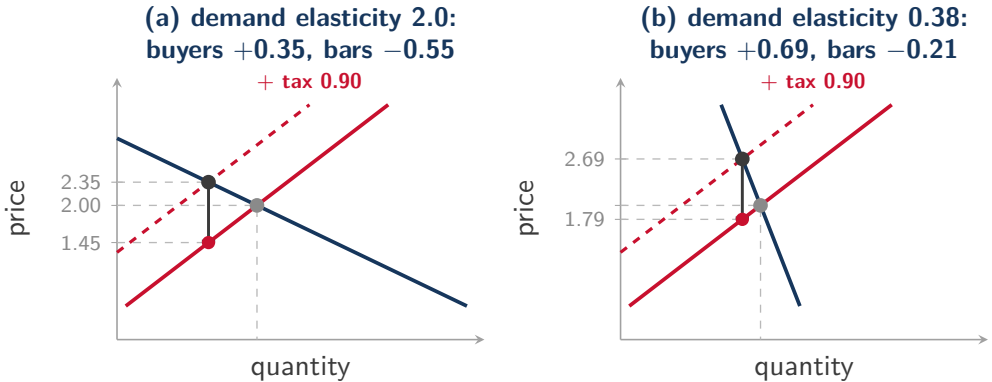
menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

1. No tax: price and menús?
2. Tax 2.50: menús, buyers pay, bars keep?
3. Tax collected a day?
4. Who bears more of the 2.50?

Class 7's markets, taxed 0.90: who cannot walk away pays more



Class 7's markets, taxed 0.90: who cannot walk away pays more



Buyers who can switch leave most of the tax to the bars; buyers who cannot pay most of it.

Tax incidence: the less elastic side bears more of the tax

	most of the tax falls on
demand less elastic	buyers
supply less elastic	sellers
equally elastic	half each

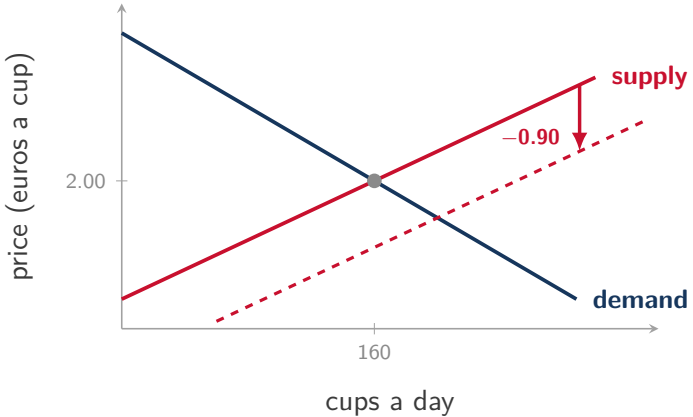
The law says who pays the tax office; elasticities say who bears the tax.

In pairs, three minutes: who bears most of the tax?

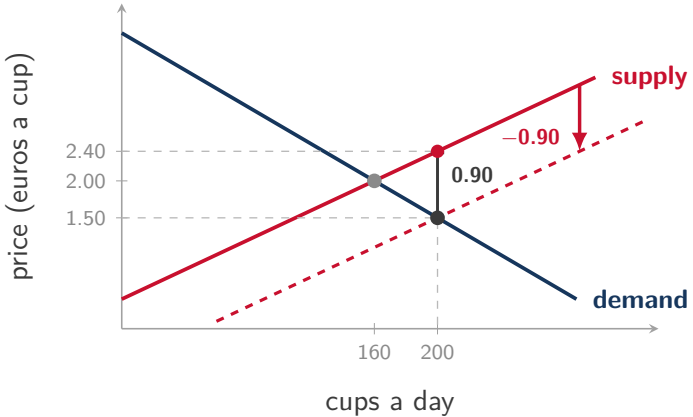
Buyers or sellers — and why?

1. Cigarettes: demand elasticity 0.4, supply 2.0
2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
3. Tax 1.00; buyers' price 5.00 $\rightarrow$ 5.80. Sellers keep? Less elastic side?
4. The same tax, paid by the buyers: the seller's price, tax not included?

A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



The tax reversed: buyers save 0.50, the bars gain 0.40.
The state pays $0.90 \times 200 = 180\,€$ a day.

Per unit or per cent: VAT is a percentage of the price

- A fixed sum per unit: a **specific tax** (0.90 a cup)
- A share of the price: **ad valorem** (VAT)
- Spain's VAT: 21 %, 10 %, 4 %
- A loaf: $1.00 + 4\,\%$ VAT $= 1.04\,€$

Either way a wedge; the elasticities decide who bears it.

Revealed in class

Before class: by slide 20's rule, who should get most of a VAT cut on bread — shoppers or shops? Bread is a necessity, and a supermarket can stock more at about the same cost.

Which curve moves? (optional)

Cinema tickets in one city

1. A tax of 1.00 a ticket, paid by the cinemas: which curve shifts?
2. Tickets get dearer: does demand shift too?
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
4. And the ticket price, and the tickets sold?

Who gains from trade? (optional)

Hours for one unit

	1 kg of cheese	1 bottle of wine
valley farm	2	1
hill farm	3	6

1. Who should make the cheese, and why?
2. At 1 bottle a kg: each farm's gain?
3. Hill farm at 4 h a kg, 2 h a bottle: any gain?

Surplus: value minus price for buyers, price minus cost for sellers.

The market's quantity makes total surplus as large as it can be.

A tax or a subsidy is a wedge; the less elastic side feels most of it.

Deadweight loss and price controls: Thursday.

Before Thursday 22 October

- Thursday: price controls; international trade and tariffs
- Read Mankiw & Taylor: Ch. 7, pp. 159–163 and 172–178; Ch. 17, pp. 380–390
- Problem set 1: due Thursday 22 October, on paper, in class
- Practise in Sextant: the units the course page lists



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